

Solution Code - Module 3-Core Java - OperatorPrecedence

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: Q08_OperatorPrecedence.java

```
1 | public class Q08_OperatorPrecedence {
2 |     public static void main(String[] args) {
3 |         // Expression 1: Multiplicative (*) has higher precedence than Additive (+)
4 |         int expression1 = 10 + 5 * 2;
5 |         System.out.println("Expression: 10 + 5 * 2");
6 |         System.out.println("Result: " + expression1);
7 |         System.out.println("Explanation: '*' has higher precedence than '+'. So, 5 * 2 = 10, then 10 + 10
= 20.");
8 |
9 |         // Expression 2: Parentheses override operator precedence
10 |        int expression2 = (10 + 5) * 2;
11 |        System.out.println("\nExpression: (10 + 5) * 2");
12 |        System.out.println("Result: " + expression2);
13 |        System.out.println("Explanation: Parentheses '()' have the highest precedence. So, (10 + 5) = 15,
then 15 * 2 = 30.");
14 |
15 |        // Expression 3: Division, multiplication, modulus, and addition
16 |        int expression3 = 100 - 20 / 4 * 3 + 15 % 4;
17 |        System.out.println("\nExpression: 100 - 20 / 4 * 3 + 15 % 4");
18 |        System.out.println("Result: " + expression3);
19 |        System.out.println("Explanation:");
20 |        System.out.println("1. '/', '*', and '%' have equal precedence and are evaluated left-to-right
(associativity):");
21 |        System.out.println("    - 20 / 4 = 5");
22 |        System.out.println("    - 5 * 3 = 15");
23 |        System.out.println("    - 15 % 4 = 3");
24 |        System.out.println("2. The expression simplifies to: 100 - 15 + 3");
25 |        System.out.println("3. '-' and '+' have equal precedence, evaluated left-to-right:");
26 |        System.out.println("    - 100 - 15 = 85");
27 |        System.out.println("    - 85 + 3 = 88");
28 |    }
29 | }
```